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Middle Permian Depositional Environments Inferred from Illite Crystallinity and Illite Chemistry Indices of X-Ray Diffraction Patterns, Saudi Arabia

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Abstract

The mineralogical composition along with the illite crystallinity and the illite chemistry indices are generally controlled by a combination of several factors such as provenance, diagenesis, and depth. These mineralogical parameters also can be used to reflect a short-term climate change. Knowing that depositional environments and sedimentary sequences are strongly affected by climate, the characterization of the Middle Permian sediments from Central Saudi Arabia is investigated using illite crystallinity and illite chemistry index to test whether this technique can be used to differentiate between different sedimentary facies. The upper Member generally consists of coastal/fluvial, fluvial, fluvial/aeolian and dune deposits representing a series of siltstones, argillaceous sandstones, and cross-bedded sheet sands with interbedded clean sandstones. The lithofacies along with the depositional environment of the aforementioned facies are readily distinguished using petrography.

The illite chemistry index is determined by the ratio of peak intensities at 5 Å to 10 Å, indicating the degree of weathering, the provenance, and a proxy for indicating climate change. Highly weathered Al-rich illite (Chemistry Index >0.5), for example can reflect a strong acidic environment where hydrolysis is prevalent, while un-weathered Fe/Mg-rich illite (Chemistry Index <0.5) can reflect a dry arid alkaline environment. The illite chemistry index of the studied samples varies from 0.51 to 0.89, with the lower values recorded for coastal/fluvial facies and the highest values for fluvial/aeolian to dune facies.

The crystal particle size of a clay mineral is generally determined by the Crystallinity degree parameter where higher degrees of crystallization are defined by a low crystallinity value. In general, the illite crystallinity is based on the measurement of the full width at half maximum (FWHM) at a 10 Å diffraction peak which can be used to classify levels of crystallinity from excellent (FWHM<0.4), to good (0.4-0.6), to moderate (0.6-0.8) and finally poor (FWHM>0.8). In this study, the values vary from 0.39 to 0.95 with lower values that are identified for fluvial/aeolian to dune facies and the highest values recorded for fluvial sedimentary facies. Low values of Illite crystallinity could be indicative of low temperature and dry climatic conditions.

Results indicate that illite crystallinity and illite chemistry indices are changing according to the depositional environments as determined by petrography. Illite crystallinity vs Illite chemistry diagram of

typical samples shows the ability to differentiate between 3 zones, including fluvial/aeolian to dune, coastal/fluvial and fluvial facies groups.

Keywords: illite crystallinity, illite chemistry, Middle Permian, Saudi Arabia, Depositional environment

Introduction

The Middle Permian sediments of Central Saudi Arabia serve as a critical record of the region's geological history. The Middle Permian Upper Member, in particular, is characterized by diverse depositional environments, including coastal/fluvial, fluvial, fluvial/aeolian and dune deposits (Figure 1; Khalifa, 2015; Soua, 2019; Polo et al., 2020). Understanding these environments is essential for reconstructing the paleogeographical and paleoclimatic conditions of the time (e.g. Soua, 2024).

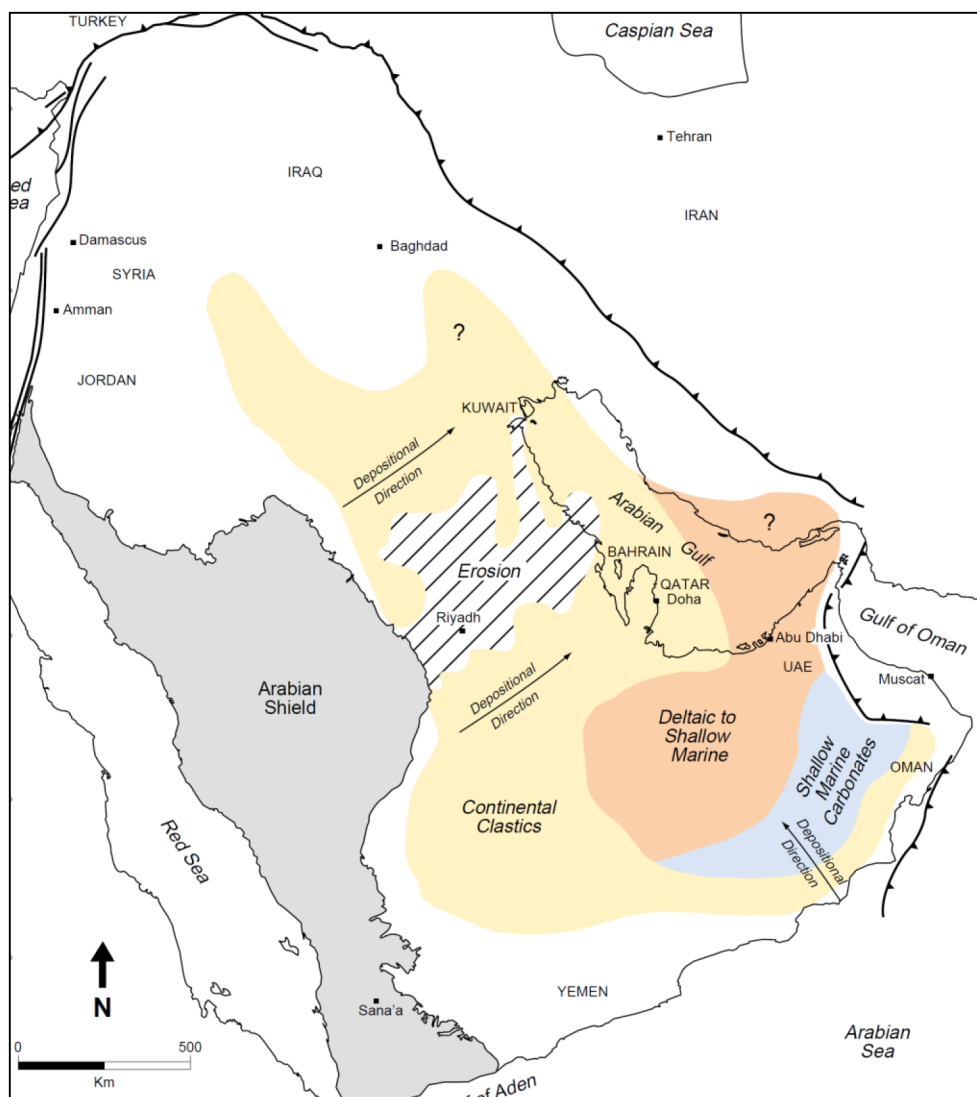


Figure 1—Early Permian: Environments of Deposition Adapted from Konert et al. (2001).

Illite, a common clay mineral that forms generally under varying environmental conditions, serves as a key indicator in sedimentological studies (e.g. Kumari and Mohan, 2021; Aboudi Mana et al., 2017; Meunier and Velde, 2013). Its crystallinity index and chemical composition provide valuable insights into the conditions of sediment formation and subsequent diagenesis (e.g. Saha et al., 2020; Kubler and Jaboyedoff, 2000; Jaboyedoff et al., 2001; Aparicio and Galan, 1999; Kish et al., 2004). This study aims

to characterize the mineralogical composition and environments of the upper Member of the Middle Permian sediments while examining the utility of illite crystallinity and chemistry indices as proxies for distinguishing sedimentary facies influenced by climatic factors.

Illite is typically identified by a set of peaks located at approximately 10 Å, 5 Å, and 3.3 Å (e.g. [Pevear, 1999](#); [Meunier and Velde, 2013](#); [Zviagina et al, 2020](#)), with the peak at 5 Å being roughly one-third the intensity of the other two (e.g. [Hillier, 1999](#)) (Figure 2). Confusion may occasionally occur due to the application of the term illite (e.g. [Meunier and Velde, 2013](#)), which refers both to a specific mineral and, in a broader context, to the more heterogeneous micaceous component found in sedimentary rocks (see [Bailey et al. 1986](#)).

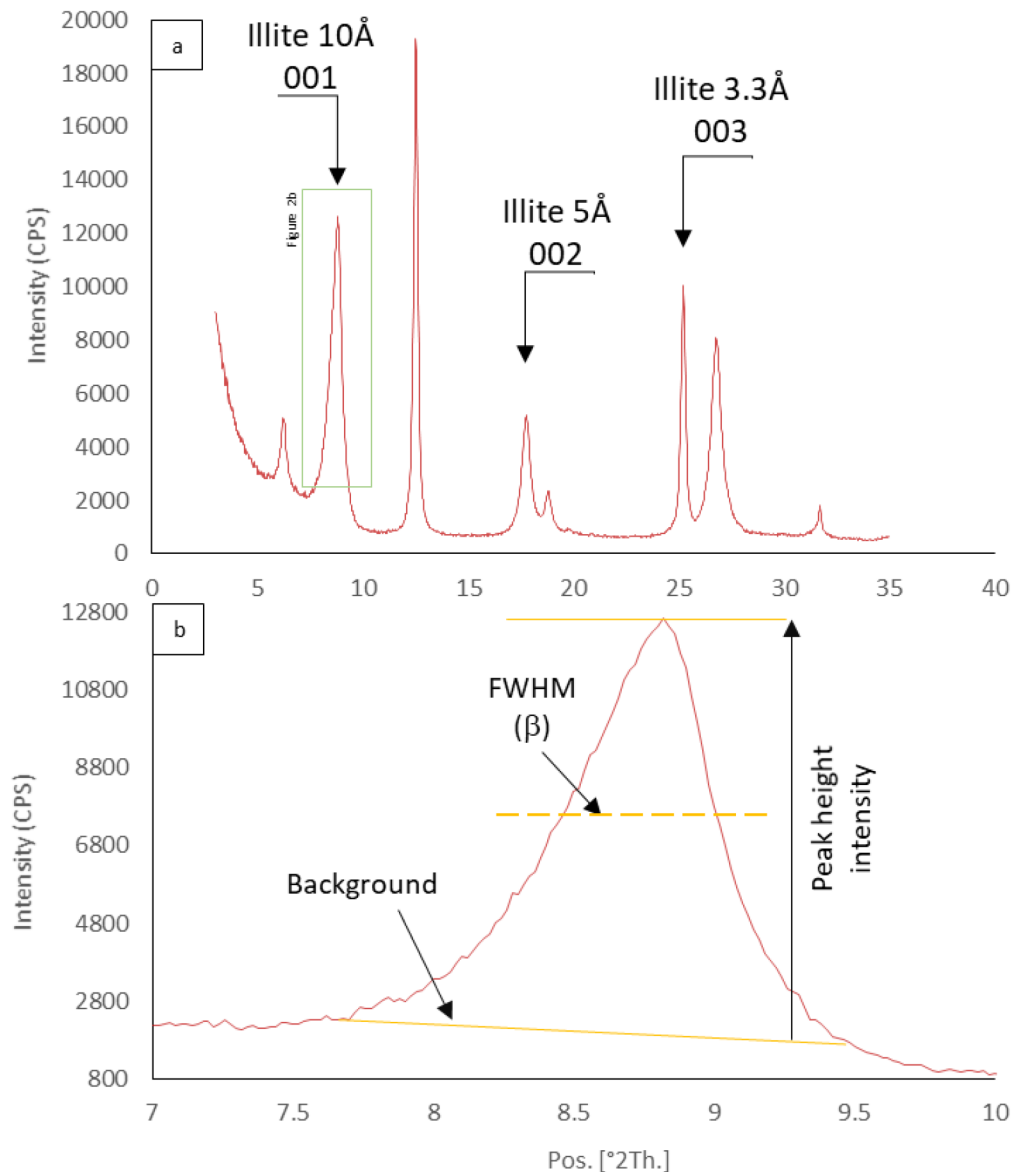


Figure 2—Typical air-dried clay pattern for the analyzed samples from Middle Permian.
 (a) definition of the main basal reflections of illite, (b) magnification of the illite 001 peak (green box in Fig. 1a) and definition of the Full Width at Half Maximum (FWHM) calculation.

Although it is hard to distinguish between illite and muscovite by XRD because they have identical crystal structures, the main difference being size of the mineral grains. Similar crystallographic planes are grouped under the name "illite/mica".

Weaver (1960) was among the initial authors to utilize the geometry of the first illite basal reflection (001 at 10Å d-spacing). However, the most renowned contribution came from Kubler (1967, 1968), after whom the illite crystallinity index was informally named. The application of this straightforward index has faced considerable debate among geologists, who have embraced and continue to utilize it, and crystallographers, who have justifiably highlighted the numerous inaccuracies that such an index conceals. Besides illite, various indices have emerged that assess the "crystallinity" of other minerals, including kaolinite (refer to the review by Galan et al., 1994) and chlorite (Arkai, 1991; Arkai et al., 1995). Subsequently, the term Kubler index became widely used in numerous publications by various authors (e.g., Eberl and Velde, 1989).

The illite crystallinity index, often known as the Kubler index (KI: Kubler, 1968; Guggenheim et al., 2002), is evaluated by measuring the width at half height of the 001 illite reflection (typically represented as the full width at half maximum (FWHM) of the illite peak at 10 Å, see Figure 2 for more details) from samples that have been air-dried and glycolated. This index offers valuable insights into the thermal history and metamorphic conditions that sedimentary rocks have undergone.

According to Abad (2007), lower KI values signify greater crystallinity, indicating that the illite has experienced more significant diagenesis or metamorphism, whereas higher KI values denote reduced crystallinity, suggesting a more recent formation or less intense metamorphic conditions (e.g., Eberl, 1993). This parameter is especially beneficial for distinguishing metamorphic zones (Blenkinsop, 1988; Arkai, 1991; Meunier and Velde, 2013). For example, research has demonstrated that illite crystallinity can effectively characterize low-grade metamorphic zones, enabling geologists to deduce the metamorphic history of a specific area (e.g., Frey and Robinson, 2009; Meunier, 2005). The correlation between illite crystallinity and metamorphic grade is well-established, with increasing crystallinity associated with elevated metamorphic temperatures and pressures (Arkai, 2002).

However, the illite chemistry, particularly its cation composition and the presence of substitutions within its structure, provides additional insights into its formation conditions and provenance (e.g., Srodon et al., 1986). Illite typically contains significant amounts of potassium (K), aluminum (Al), and silicon (Si), with variations in these elements reflecting the source material and environmental conditions during formation (Meunier and Velde, 2013; Barre et al., 2007; Weaver, 1967). The chemical composition can also indicate the degree of weathering and the influence of diagenetic processes.

Illite chemistry is often analyzed as the ratio of illite 5A over 10A peak height intensities (Figure 2; Choudhary et al., 2025; Song and Sim, 2021). Variations in the chemical composition of illite can be linked to specific depositional environments, such as fluvial, interdune or dune settings, providing valuable information for reconstructing past geological conditions (e.g., Velde et al., 1986; Curtis, 1990; Deconinck et al., 2003).

Depositional environments of siliciclastic sediments are in general determined using petrographic analysis performed on sandstone thin section samples. This is considered the most reliable technique to relate grain size to depositional environment. In addition to this, change in clay content and crystallinity index of illite present in the fraction can be linked to the energy conditions and, hence, depositional environment (e.g., Xu et al., 2017; Grathoff et al., 2001). In addition, petrography can be used to determine the provenance, which can also influence the peak height intensities produced by X-ray diffraction.

In this study, the crystallographic planes 001 and 002 of illite were utilized for calculating the related crystallinity and chemistry indices (Figure 2). However, one must consider also checking the other intensities at a higher angle from the diffraction pattern to check if these differ. If there is a significant difference between them, then this could signal that some of the intensities could result from preferred orientation during sample preparation.

The main purpose of this paper is to use the illite crystallinity and illite chemistry indices as a discriminator based on the assumptions that the related physical meaning is related to the energy and provenance of the depositional environment, and that Illite is the dominant clay mineral of the clay fraction of the study samples.

Geological overview

The samples utilized in this paper have been interpreted as having been deposited in a mixed coastal/fluvial, fluvial, fluvial/aeolian and dune environment, situated within an arid to semi-arid depositional context (Al-Masrahy, 2020).

Various continental facies types are observed, including eolian ergs, ephemeral streams, and ephemeral playa systems (refer to Soua, 2019 and the references therein for additional details).

Recent findings by Polo et al. (2019) have reported evidence of marine sedimentation, specifically in the form of marginal marine, tidally-influenced deposits.

The Middle Permian sandstones found in the study area consist of fine to medium-grained quartz-rich sandstones (see Figure 1). These sandstones represent sequences that include coastal/fluvial, fluvial, fluvial/aeolian and dune deposits, characterized by cross-bedded sheet sands interspersed with clean sandstones, argillaceous sandstones, and siltstones (Al-Masrahy, 2020).

Materials and Methods

Sample Collection and Preparation

Core samples were systematically collected from different facies within the upper Member of the Middle Permian formation. The selected facies included coastal/fluvial, fluvial, fluvial/aeolian and dune deposits. Each sample was meticulously prepared by fine grinding to ensure uniformity for X-ray diffraction analysis, which is crucial for obtaining accurate mineralogical data.

X-ray Diffraction Analysis

Mineralogical analysis was carried out on randomly oriented powders by using a Rikagu ULTIMA IV powder X-Ray diffractometer with copper (Cuka) radiation (40 kV, 40 mA), in the 3°-70° (2 θ) interval with a step size of 0.02° increment and a speed of scan of 10.

Interpretation of the XRD patterns was made with High Score Plus software, combined with the International Powder Diffraction Data (ICDD) of the powder diffraction file (PDF-4+ Minerals) database of the standard reference materials using Rietveld refinement procedure at Exploration Core Laboratory (ECLD).

The clay fraction procedure is employed for the determination of crystalline clay size (<2 microns equivalent spherical diameter) components. Besides the clay slide being analyzed in an air-dried state, the sample is also exposed to Ethylene glycol fumes, which causes the spacing of the basal planes to increase for expandable clay minerals. This allows easy differentiation between illite, smectite, and ordered and randomly interstratified clay minerals. For a properly prepared sample, the results are highly reproducible. However, the accuracy of the results can vary greatly dependent on the specific composition of the sample, its homogeneity and its crystallinity. Analysis was carried out on randomly oriented powders in the 3°- 35° (2 θ) interval with a step size of 0.02° increment and scan speed of 2. Quantification was carried out using the Mineral Intensity Factor method.

The crystallinity of illite was examined using the aforementioned X-ray diffraction setup. Illite crystallinity (IC) was assessed by measuring the full width at half maximum (FWHM) of the 10 Å diffraction peak, providing insights into the thermal history and diagenetic processes that affected the sediments. The illite chemistry index (ICI) was calculated by determining the ratio of peak intensities at 5 Å to 10 Å. This index serves as a measure of the degree of weathering and provenance.

Results

Mineralogical Composition

The Middle Permian exhibits a diverse mineralogical composition that varies significantly across the different analyzed wells. Fluvial deposits are primarily composed of siltstones and argillaceous sandstones, characterized by high variability in grain size and mineral content. In contrast, coastal/fluvial facies are dominated by fine-grained sediments that exhibit lower weathering indices, indicative of a more stable depositional environment.

Coastal dune and aeolian facies, on the other hand, are characterized by cross-bedded and flat-laminated quartz arenites. These facies display higher weathering indices and distinct mineralogical signatures, reflecting the dynamic conditions present during their formation.

High percentage of K-feldspar and clay minerals were also identified in few intervals indicating the proximal or fluvial affinity where the wind action has reworked the feldspar, transforming it into clay in parts. This transformation has in turn enhanced the reservoir quality within the aeolian dune facies association.

XRD analyses of bulk samples and clay fractions from the Middle Permian sandstones reveal significant variations in their mineralogical compositions. The Middle Permian consist mainly of quartz and phyllosilicate minerals. The phyllosilicate group includes illite, kaolinite, mixed-layered illite-smectite (I-S), chlorite, with no discrete smectite is being identified. Additionally, trace amounts of hematite, gypsum, barite, siderite were also identified in the analyzed rock samples (Soua, 2024) along with minor amounts of potassium feldspar (microcline), plagioclase (albite), dolomite, anhydrite, and halite.

Clay minerals from the clay fraction proportion are detailed in this study (Figure 3).

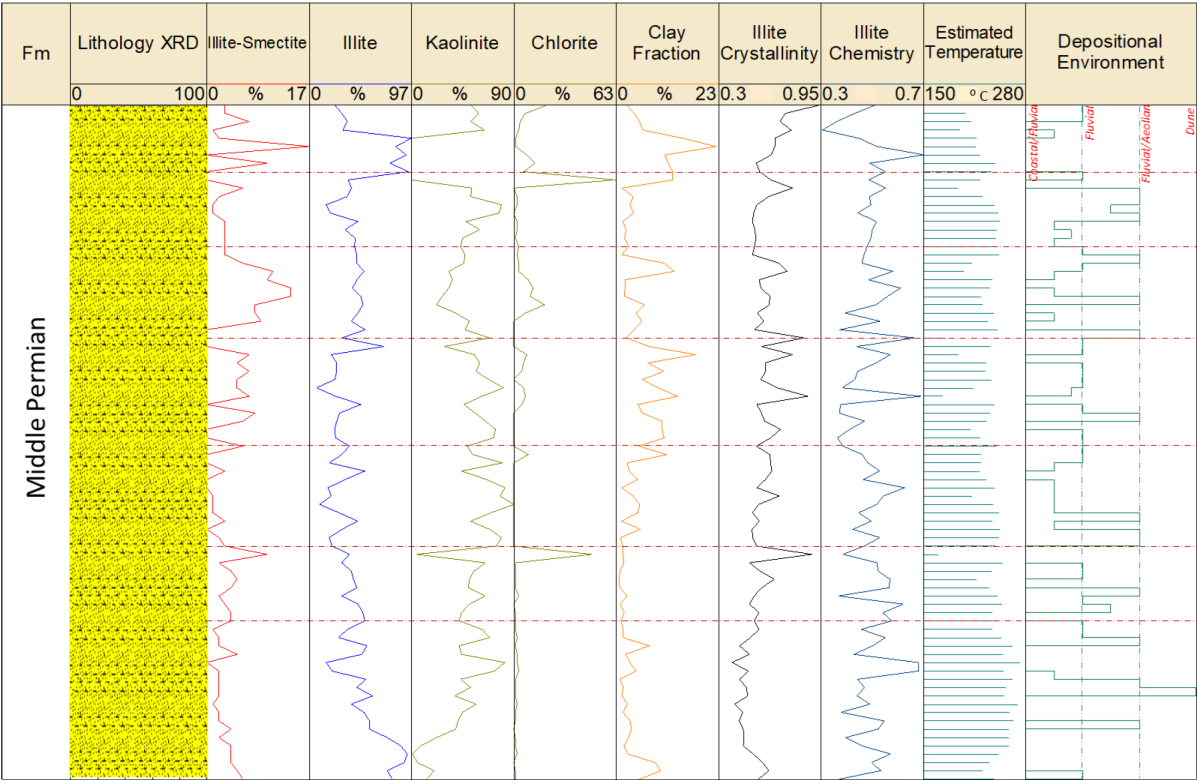


Figure 3—Summary of clay identification and quantification of main parameters discussed in this paper (illite crystallinity, Illite chemistry, estimated temperature from illite crystallinity and inferred depositional model). Horizontal dashed red lines indicate the change of well, blue vertical dashed lines indicate the change in depositional environment. The temperatures were estimated using the following formula: $T\text{ }^{\circ}\text{C}=353-206\cdot\text{FWHM}\text{ (}\text{\AA}^{\circ}20\text{)}$ (Mukoyoshi et al. 2007; Fukuchi et al., 2014). Fm: Formation

Figure 3 details the vertical variations of samples taken from 7 different wells separated by red dashed lines. The distribution of the mineral variation is not in-depth scale. The mixed layered I/S is observed predominantly in significant values in only two wells (Figure 3), while in the other six wells, it is found in a low proportion. Conversely, illite and kaolinite are the major component of the clay fraction.

Illite Chemistry Index

The illite chemistry index (ICI) values observed in the study range from 0.3 to 0.7. Notably, lower ICI values were associated with coastal/fluvial facies, suggesting reduced weathering and a more subdued climatic influence. Conversely, higher ICI values were recorded in fluvial/aeolian to dune deposits, indicating increased weathering and a more active climatic regime.

Figure 4 shows that the illite chemistry index is mostly above 0.45, suggesting that illite is abundant in the Aluminum (Al) element, while some of the samples plot within the Fe-Mg field indicating species such as biotite and mica. It appears that most of the illite primarily result as illitization most probably from kaolinite.

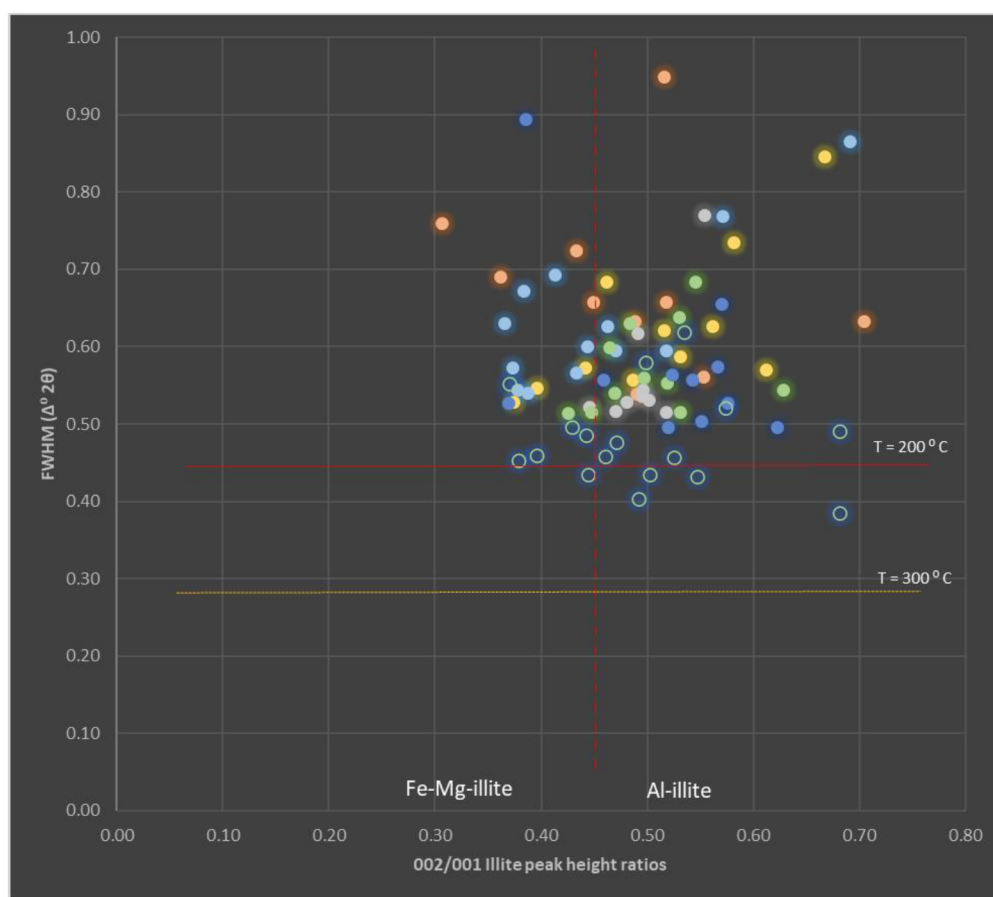


Figure 4—illite chemistry versus illite crystallinity indices distribution in the different analyzed wells

Illite Crystallinity Measurements

Illite crystallinity values varied from 0.38 to 0.95°A20 across the sampled facies. Dune and fluvial/aeolian facies exhibited lower FWHM values, suggesting higher crystallinity, which may indicate a more stable thermal history. In contrast, fluvial deposits displayed higher FWHM values, reflecting a more dynamic depositional environment characterized by fluctuations in temperature and moisture. Overall, the illite crystallinity can be classified as good from the range point of view. As illustrated in Figure 4, illite chemistry and illite crystallinity display contrasting trends with generation is being most likely happening in relatively low temperatures.

Discussion

Background

The crystal particle size of a clay mineral is generally determined by the Crystallinity degree parameter where higher degrees of crystallization are defined by a low crystallinity value. Although the crystallinity index is influenced by various factors, it has gained widespread acceptance, leading to the publication of numerous scientific articles (Kisch, 1983; Frey, 1987; Weaver, 1989; Frey and Robinson, 2009). The measurement of the index is fundamentally linked to the thickness of illite particles, and therefore it quantifies the full width at half maximum intensity (FWHM) of the first X-ray powder-diffraction peak of dioctahedral illite-mica at 10 Å (referred to as the Scherrer width) (e.g. Drits et al., 1997; Eberl et al., 1998; Abad, 2007). While this factor is not the sole one, it is a significant element that influences the width of the X-ray diffraction reflection. Additional factors include: the effects of the instrument (its calibration, Lorentz polarization factor, etc.), variability in chemical composition (X-ray diffraction structure factor), the presence of expanding interlayers, defects within the crystal structure (mainly Illite-Smectite mixed layers), and the method of sample preparation (e.g. Klug and Alexander, 1974; Eberl and Blum, 1993; Moore and Reynolds, 1997). Furthermore, the lithology of the sample itself is also a significant factor. Another problem of the geological interpretation of illite crystallinity index is their use in sedimentary rock containing illite-smectite with high expandability, which may affect the shape and broadening of the 001 basal reflection of illite and therefore the FWHM value will be considered as an artifact rather than a true value. In the examined samples, the I/S mixed layers are minimal, and no expandability was observed.

The Kubler index, initially introduced by Kubler (1964) and further developed by Kubler (1967 and 1968), as well as by Dunoyer de Segonzac et al. (1968). This measurement is conducted on the <2gm size-fraction of air-dried clay samples utilizing Cu-K α radiation. This index, represented as minor variations in the Bragg angle ($A^\circ 2\theta$), was established as a technique for delineating the diagenesis- anchizone and anchizone-epizone metamorphic boundaries (Kisch, 1990, 1991; Warr and Rice, 1994; Merriman et al., 1995; Warr and Nieto, 1998; Abad et al., 2001; Kisch et al., 2004).

Illite represents the initial phase that generates a 10 Å peak in the smectite-illite/smectite-illite-mica reaction series. Therefore, a specific crystallinity value is dependent on the reaction progress of dioctahedral phyllosilicates found in low-grade rocks, reflecting a state influenced by kinetic factors whenever the smectite-to-illite reaction has advanced to over 90% illite (e.g. Abad, 2007). Essene and Peacor (1995) emphasize that when these geological variables are roughly equivalent, a strong correlation should exist between reaction progress (e.g. as measured by KI) and metamorphic grade (temperature).

Illite as an Indicator of Depositional Environment

The illite crystallinity and the illite chemistry indices have been extensively utilized to signify significant weathering (Wan et al., 2012; Wang and Yang, 2013).

In general, an illite chemistry index of below 0.45 indicates Fe-Mg-rich illite, which is a result of physical erosion, whereas an illite chemistry index exceeding this number signifies Al-rich illite, indicative of strong hydrolysis.

A higher illite crystallinity suggests lower crystallinity, which may point to significant hydrolysis in the terrestrial provenance area under dry and cold climatic conditions (e.g. Hu et al., 2014).

This may explain somehow the low temperatures in which the illite was generated along. This explains also the origin which may be kaolinite illitization.

The results of this study underscore the pivotal role of illite characteristics in distinguishing between various depositional environments. The illite chemistry index serves as a reliable proxy for assessing weathering intensity and climatic conditions, while illite crystallinity provides insights into the thermal and diagenetic history of the sediments.

The temperatures were estimated using illite chemistry vs. illite crystallinity index diagrams and also using the following formula:

formula: $T\ (^{\circ}\text{C}) = 353 - 206\text{FWHM}\ (\text{\AA}^{\circ}2\theta)$ (Mukoyoshi et al. 2007; Fukuchi et al., 2014), which gives a slight difference.

The estimated paleotemperatures are 157°C to 273°C. The relatively lower temperatures estimated are associated with coastal/fluvial to fluvial facies while higher temperatures are linked to fluvial aeolian and dune facies (see Figure 3).

The bulk XRD results show that plagioclase (chiefly albite) is very low to absent in all analyzed samples. Hemley and Jones (1964) illustrated that albite converts to illite due to alteration occurring under the following condition:



In addition, the high presence of kaolinite in the clay fraction may suggest that part of the illite is a product of kaolinite illitization (see Mantovani and Becerr, 2010; Bello et al., 2025; Berger et al., 1997).

The high recorded values of Illite Crystallinity are particularly roughly associated with the Coastal/Fluvial facies (Figure 5) indicating strong hydrolysis process, and therefore may possibly indicate hydrothermal alteration that produced large amounts of illite. Coastal/Fluvial facies is roughly associated with high illite crystallinity index along with higher illite chemistry values, while Fluvial facies is linked to lower illite chemistry and moderate to high illite crystallinity. Fluvial/Aeolian to Dune facies is roughly associated with low illite crystallinity and moderate illite chemistry (Figure 5).

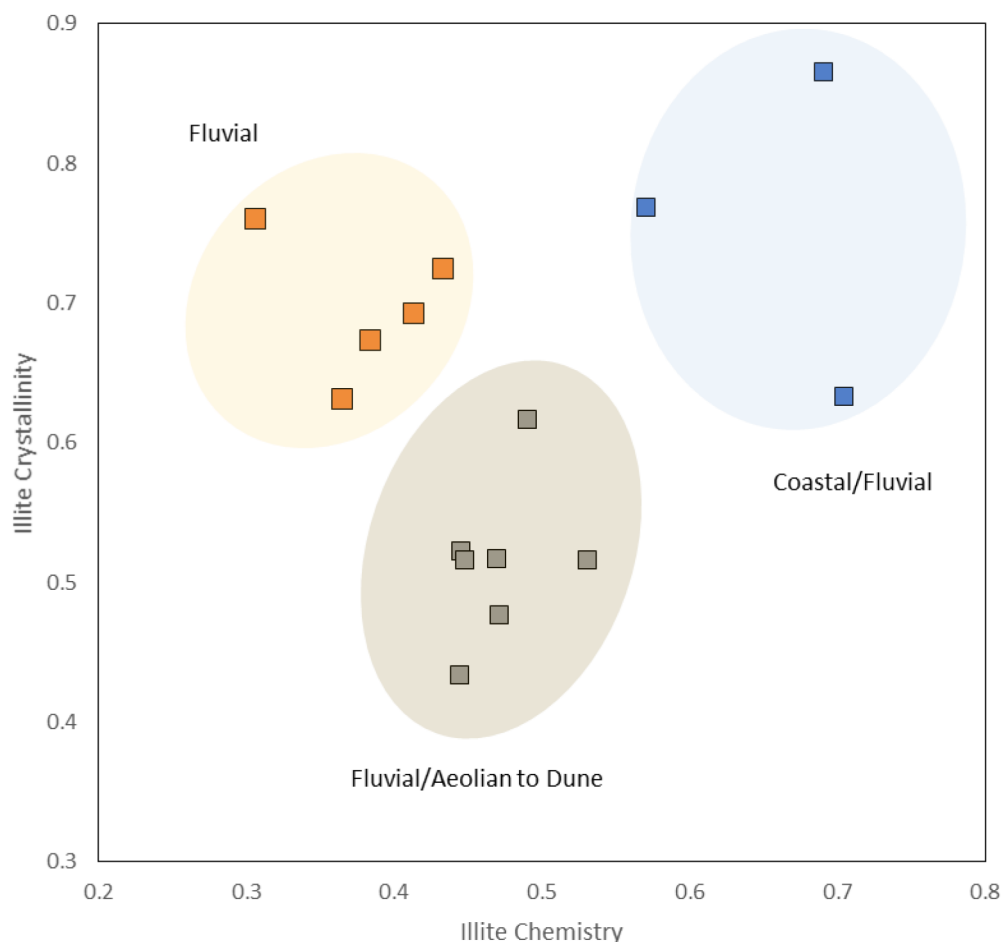


Figure 5—Selected samples and differentiation of depositional environments using illite crystallinity versus illite chemistry

In general, the illite crystallinity is based on the measurement of the FWHM at a 10 Å diffraction peak which can be used to classify levels of crystallinity from excellent (FWHM<0.4), to good (0.4-0.6), to moderate (0.6-0.8) and finally poor (FWHM>0.8). In this study, the values vary from 0.39 to 0.95 with lower values that are identified for fluvial/aeolian to dune facies and the highest values recorded for Coastal/Fluvial and fluvial sedimentary facies. Low values of Illite crystallinity could be indicative of low temperature and dry climatic conditions.

Weathering and Climatic Conditions

The observed trends in illite chemistry and crystallinity highlight the impact of climatic factors on sedimentary processes. Higher ICI values in coastal and aeolian facies suggest a more pronounced weathering regime, likely associated with increased rainfall and temperature fluctuations. Conversely, the lower ICI values in lacustrine deposits indicate a more stable, less weathered environment, which is consistent with the conditions expected in such settings.

The illite chemistry index is determined by the ratio of peak intensities at 5 Å to 10 Å, indicating the degree of weathering, the provenance, and a proxy for indicating climate change. Highly weathered Al-rich illite (Chemistry Index >0.5), for example can reflect a strong acidic environment where hydrolysis is prevalent, while un-weathered Fe/Mg-rich illite (Chemistry Index <0.5) can reflect a dry arid alkaline environment. The illite chemistry index of the studied samples varies from 0.51 to 0.89, with the lower values recorded for Fluvial facies and the highest values for Coastal/Fluvial facies (Figure 5).

Table 1 summarizes the main characteristics of Illite chemistry and illite crystallinity indices.

Table 1—Summary of the main aspects describing the illite chemistry and illite crystallinity indices.

Parameter	0 to 0.5	0.5 to 1	Comments
Illite chemistry index	Chemistry Index <0.5: un-weathered Fe/Mg-rich illite reflecting a dry arid alkaline environment.	Chemistry Index >0.5: highly weathered Al-rich illite, reflect a strong acidic environment where hydrolysis is prevalent.	It is determined by the ratio of peak intensities at 5 Å to 10 Å, indicating the degree of weathering, the provenance, and a proxy for indicating climate change.
Illite crystallinity index	excellent crystallinity (FWHM<0.4), to good crystallinity (0.4-0.6).	Moderate crystallinity (0.6-0.8) to poor crystallinity (FWHM>0.8).	It is based on the measurement of the FWHM at a 10 Å diffraction peak which can be used to classify levels of crystallinity.

The findings of this research contribute to a more nuanced understanding of the geological history of the Middle Permian sandstone and the origin of the clay association including illite and kaolinite. By establishing clear links between illite characteristics and depositional environments, this study enhances our ability to reconstruct past climatic conditions and sedimentary processes.

The estimated temperatures from the Illite Crystallinity Index show relatively lower values which corroborate in some circumstances with the overall Middle Permian paleoclimate.

Conclusions

In conclusion, this study demonstrates that the mineralogical composition, Illite Crystallinity (Kubler Index) and Illite Chemistry indices of the Middle Permian sediments in Central Saudi Arabia are roughly linked to a variety of geological and climatic factors, including provenance, diagenesis and burial variations. The findings reveal that these parameters serve not only as indicators of sedimentary facies but also as proxies for short-term climate changes.

The analysis of sedimentary environments has successfully classified the Upper member into distinct facies, each reflecting unique depositional conditions. The observed variations in illite chemistry indices and crystallinity values provide critical insights into the extent of weathering and climatic influences,

highlighting the contrast between more acidic and arid environments. Specifically, the differentiation of Fluvial/Aeolian to Dune facies from Coastal/Fluvial deposits underscores the potential of these mineralogical tools in reconstructing past environmental conditions.

Future studies should expand on these findings, exploring additional parameters and refining the ones used in this study.

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